MUHAMMAD ZUL REZKY ZAAKI

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ABOUT ME

IT fresh graduate (GPA 3.73, Cum Laude) from Telkom University with hands-on experience across full-stack development, machine learning, and data processing, including a Full-Stack Developer internship at IPDN and a published IEEE conference paper on deep learning-based waste classification. Proficient in Python, PHP, and JavaScript with frameworks such as Laravel, React.js, and Node.js. Complementing my technical background, I bring proven leadership experience as Vice President of a 500+ member student organization, sharpening my skills in team management and cross-divisional coordination. Eager to contribute to a software engineering team with a strong drive to learn and deliver real impact.

EDUCATION

Telkom University – Bandung, Indonesia **September 2022 – August 2026**

Bachelor of Informatics

- GPA : 3.73/4.00 — *Cum laude*
- Publication : IEEE Conference Paper "Organic and Inorganic Waste Detection using SwinConvNeXt Fusion Model" — AIMS 2026, Bandung (Hybrid), 24 July 2026. Co-authored with Putu Harry Gunawan.

PROFESSIONAL EXPERIENCE

Institut Pemerintahan Dalam Negeri (IPDN)

Full-Stack Developer Intern **July 2025 – September 2025**

- Designed an end-to-end UI/UX for IPDN's Learning Management System (LMS), streamlining the digital learning experience for hundreds of cadets.
- Developed a Telegram chatbot to deliver important information to praja (cadets).
- Performed sentiment screening and classification of 350+ comments on IPDN's Instagram account, categorizing them into positive and negative sentiment.
- Built a face recognition web app that identifies cadets from a photo and matches their name and NPP.

PROJECT EXPERIENCE

Organic & Inorganic Waste Classification using SwinConvNeXt Fusion Model

Final Thesis Project **February 2026 – June 2026**

- Designed SwinConvNeXt, a hybrid architecture combining Swin Transformer and ConvNeXt in a parallel-branch design with a fusion layer, built using PyTorch and timm.
- Trained on a 15,515-image dataset with severe class imbalance (89% vs. 11%), applying weighted loss and oversampling to improve minority-class detection.
- Achieved 98.75% accuracy, 98.77% F1-Score, and 99.74% ROC-AUC, with stronger organic-waste recall (96.43%) than the higher-accuracy EfficientNet-B0 baseline.
- Validated results using Grad-CAM and t-SNE visualizations to confirm the model's feature-learning quality.

Autonomous Tank Robot

Team Project – 5 Members **September 2025 – December 2025**

- Built dual-controller embedded architecture (Raspberry Pi + Arduino) handling LIDAR-based obstacle detection and real-time motor control.
- Trained a YOLOv8-Nano object detection model, optimized with TensorFlow Lite for real-time inference on edge hardware.
- Developed a live web control dashboard using Flask + WebSocket for real-time video streaming and gamepad teleoperation.

IPDN Cadet Face Recognition System

Internship Project – IPDN **July 2025 – September 2025**

- Built a face recognition pipeline using MTCNN for face detection and alignment, FaceNet (InceptionResnetV1 pretrained on VGGFace2) for 512-dimensional embeddings, and a linear SVM to classify cadet identity.
- Applied data augmentation with Albumentations (brightness/contrast, flip, shift-scale-rotate) to keep the model robust with only a few photos per cadet.
- Developed a web app with a native PHP frontend and Python backend, featuring admin login, a dataset management page to register cadets (name, NPP, photos), and one-click model retraining.
- Accurately identified 10+ cadet faces during testing, and safely returned "Not recognized" for faces outside the dataset instead of forcing a wrong match.

Praja Assistant Chatbot with n8n AI Agent

Internship Project – IPDN **July 2025**

- Built a Telegram chatbot for IPDN cadets using an n8n AI agent powered by Google Gemini with conversation memory, allowing cadets to make requests in natural language instead of searching spreadsheets manually.
- Integrated 7 tools (Google Sheets, Gmail, Google Calendar, and Date & Time), enabling the agent to look up a cadet's class by NPP, retrieve learning resources, send broadcast emails, and create calendar events from a single message.
- Self-hosted n8n in a Docker container and tested the bot end-to-end through live Telegram conversations.

ORGANIZATIONAL EXPERIENCE

UKM (Student Activity Unit) - Telkom University Badminton Club

Vice President **January 2025 – January 2026**

- Managed and coordinated 500+ members with 6 divisions within 60 board members, ensuring smooth operation with efficient organization administration.
- Performed the Chairperson's duties in their absence, including in meetings, official events, and other activities.
- Oversaw internal and external organization operations, training programs, logistics, design, and entrepreneurship.

Internal Coordinator **January 2024 – November 2024**

- Coordinated members of the Internal division to ensure their job responsibilities were carried out properly.
- Managed and oversaw the implementation of UKM Telkom University Badminton Club's work programs.

Event Staff **August 2023 – November 2023**

- Designed and executed activities to select 500+ prospective members for UKM.
- Responsible for determining assessment criteria including technical, physical, and interview test.

CERTIFICATES

LKMM Tingkat Dasar (LKMM-TD) – Telkom University **10 Feb 2025 – 11 Feb 2025**
LKMM Tingkat Menengah (LKMM-TM) – Telkom University **12 Feb 2025 – 14 Feb 2025**
Azure AI Fundamentals (AI-900) – Microsoft **May 2025**

SKILLS

Programming Languages: Python, PHP, JavaScript

Frameworks/Libraries: Laravel, React.js, Node.js, Flask, PyTorch, timm, TensorFlow Lite, YOLOv8

Databases: MySQL, MongoDB

Tools: Git/GitHub, Docker, n8n, Google Colab, Vercel

Soft Skills: Analytical Thinking, Problem Solving, Communication, Team Collaboration

Languages: Bahasa Indonesia (Native), English (Intermediate)